

Course Project: Guide

Machine Learning Algorithms, Weeks 4–10 and the period before the Final

You will work in pairs on one project for the rest of the course. You collect your own data, build models on it and improve them as the course moves on. The project is assessed four times, and each stage builds on the previous one.

Stage	When	What you add
Registration	by Saturday 3 October, 23:59	Team, direction, question and target, data source
Midterm	Week 5	Your data, the Week 3–4 models, results and a plan (50% of attestation 1)
Endterm	Week 9	Ensembles and tuning, an unsupervised analysis, a neural network (50% of attestation 2)
Pre-defense	Week 10	Final results and a draft of the report. Not graded, but required. You get a list of fixes
Final	Examination period after Week 10	Text or image features, a local app with your model, a report and a defense

This guide explains the steps up to the Midterm. The Midterm rubric describes how the work is graded. The Endterm and Final briefs will follow later.

1. Form a team

Teams are pairs from the same practice group. If your group has an odd number of students, one team of three is possible with the teacher's approval. Both partners are responsible for the whole project and both answer questions at every defense.

2. Choose a direction and a topic

A direction is a subject area, such as used cars or IT job vacancies. The appendix lists 48 directions with example targets. You can also propose your own direction, and the teacher approves it against the data criteria below.

A topic is your own question and target inside a direction. At most two teams across all groups can work in the same direction. If two teams share one, one of them solves a regression task and the other a classification task, and their questions differ.

Your data must meet five criteria:

Data criterion
You collect the data with your own code (web scraping or API requests) and keep the raw snapshot, the code and the date of collection.
At least 1,000 rows.
Structured features (numbers, categories) and at least one text field or images. You will use the text or images at the Final.
A target for regression or classification that the features can reasonably explain.
The data describe objects (listings, products, games, courses and similar), and the source's terms allow collecting them.

Check the source before you register: open its robots.txt and terms of use, and make sure you can get 1,000 rows without logging in. Choosing a source that works saves you a week later.

3. Register in Moodle

Open the course page, find the Database "Course Project Registration" in the General section and press "Add entry". One partner submits one entry for the whole team. The fields are:

Field	What to write	Example
Group	Your academic group	SE-2417

Team members	Full names, one per line	Aigerim Seitkali Timur Akhmetov
Direction	Number and name from the list, or Own direction	7. Used cars
Own direction	Only if you chose Own direction	
Task type	Regression or classification	Regression
Target	What your model predicts	Mileage
Question	Your question, why it matters and who would use the answer, in 3–4 sentences	Is the mileage in used-car ads plausible for the car's age, brand and price? A buyer could spot suspicious ads before calling the seller.
Main source	Link to the website or API you collect from	https://example-cars.kz
Additional sources	Other sources, if you combine several	
Text or images	What your data contain besides structured features	Text field, Images
Expected rows	How many rows you expect	3000
Goals	Minimum goal (what you will certainly deliver) and ideal goal	Minimum: 1,500 ads and three models that beat the baseline. Ideal: 5,000 ads from two sites.
Source check	Confirm that you checked robots.txt and the terms	Ticked

After you submit, the teacher reviews the entry. You can edit it until it is approved. Approved entries become visible to your group, so check them before you choose. If your direction is already full, you will be asked to pick another one. Registration closes on Saturday 3 October at 23:59.

4. Prepare the Midterm

The Midterm rubric lists what is graded. Four things need a short explanation here.

The data card is a short section at the top of your notebook or a separate file. It gives the source and its link, what one row describes, a table of all fields with their type and meaning, how and when you collected the data, how many rows you have before and after cleaning, and how you checked the source.

The contributions section says who did what, for example "Aigerim wrote the scraper and the data card. Timur built the pipeline and the models. We wrote the conclusions together." In the same place, list the AI tools you used and what you used them for. If you used none, say so.

Keep your train/test split fixed with a seed. You will reuse the same split at the Endterm and the Final, so your results stay comparable.

Slides: about 7–9 of them are enough for 7–10 minutes. A structure that works is question, data, preparation, models, results, errors, conclusions and plan. Show figures and tables, not code.

5. Submit and defend

Both partners upload the same files to the Midterm assignment in Moodle by 23:59 on the day before your defense: the notebook, the data collection code, the raw and cleaned data (or a link if the files are large), and the slides. Anything you show in the presentation has to be in the upload.

Defenses take place during your group's classes in Week 5. The teacher sets the order. Each team presents for 7–10 minutes and then answers questions for 3–5 minutes. Questions can go to either partner and about any part of the project. Come with your slides and your notebook ready to open.

After the Midterm you get your grade and short feedback. Use the feedback and your own plan for the Endterm, which applies the Week 5–8 topics to the same data. The Endterm brief comes out in Week 6.

Appendix: directions

Targets are examples. You may choose another target in the same direction. The same list is in the Moodle registration form.

Direction	Regression	Classification	Text / images for the Final	Direction	Regression	Classification	Text / images for the Final
1. Apartments for sale	price per m ²	building type (panel, brick, monolith)	description, photos	25. Eyewear	price	frame shape	description, photos
2. Long-term apartment rentals	monthly rent	renovated or not	description, photos	26. Cosmetics and skincare	average user rating	product category	description, photos
3. New residential complexes	price per m ²	housing class (economy, comfort, business)	description, photos	27. Perfume	longevity rating	fragrance family from notes	description, bottle photos
4. Houses and land plots	price	house or land-only listing	description, photos	28. Construction sets and toys	number of pieces	age group	description, photos
5. Commercial real estate	price per m ²	property type (office, retail, warehouse)	description, photos	29. Houseplants and garden plants	mature height	light requirement	care description, photos
6. Hotels	guest rating	star class	description, photos	30. Musical instruments	price	instrument category	description, photos
7. Used cars	mileage	body type	description, photos	31. Recipes	cooking time	difficulty level	ingredients and steps, photos
8. New-car catalogues	fuel consumption	vehicle segment	description, photos	32. Restaurants and cafés	venue rating	price category	venue description, photos
9. Electric vehicles	driving range	vehicle segment	description, photos	33. Wine and spirits	rating	wine style	description, label photos
10. Car rental offers	daily price	car class	description, photos	34. Craft beer	alcohol by volume	beer style	description, label photos
11. Trucks and special machinery	price	vehicle type	description, photos	35. Specialty coffee	cupping score	roast level	tasting notes, package photos
12. Motorcycles and scooters	engine power	motorcycle type	description, photos	36. IT job vacancies	salary	seniority level (junior, middle, senior)	vacancy text
13. Bicycles and e-bikes	weight	bicycle type (road, mountain, city)	description, photos	37. Non-IT job vacancies	required years of experience	remote or on-site	vacancy text
14. Laptops	battery life	segment (gaming, business, budget) without using price	title, photos	38. Online courses	course rating	course level	course description
15. Smartphones	benchmark score	price segment from specifications only	title, photos	39. Films	rating	age rating	plot summary, posters
16. Used phones and electronics	price	device condition	description, photos	40. TV series	average episode rating	highly rated or not	synopsis, posters
17. TVs and monitors	power consumption	panel technology	title, photos	41. Books	rating	fiction or non-fiction	annotation, covers
18. Home appliances	annual energy use	energy class	title, photos	42. Video games	time to complete	review class (positive, mixed, negative)	description, screenshots
19. Headphones and speakers	battery life	wired or wireless	title, photos	43. Board games	complexity weight	player-count category	description, box images
20. Mobile games and apps	user rating	app category	description, screenshots	44. Anime and manga	score	format (TV, film, OVA)	synopsis, posters
21. Furniture	price	room category	description, photos	45. Collectible card games	card price	rarity	card text, card images
22. Watches	price	mechanical or quartz	description, photos	46. Travel tours and packages	tour price	meal plan	hotel description, photos
23. Sneakers	resale price	shoe category (running, lifestyle, basketball)	description, photos	47. Tourist attractions	visitor rating	attraction type	description, photos
24. Bags and backpacks	capacity in litres	bag type	description, photos	48. Hiking trails	elevation gain	difficulty level	route description, photos

Own direction: proposed by the team and approved by the teacher.